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Protein 3D structure of benthic Foraminifera (*Ammonia dentata*), parts of East coast of India: A potential proxy for detecting marine pollution?

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A B S T R A C T

Benthic Foraminifera are known to survive and thrive in contaminated marine ecosystems where they constitute a lion's share of the benthic biomass. However, the morphological adaptations of Foraminifera to the polluted environment remain poorly constrained. We sampled from marine sediments of the Nellore region, East coast of India. The species *Ammonia dentate* (A.*dentata*) dominated the Foraminiferal biota, and metatranscriptomics generated were employed to characterize Foraminifera metabolic activity across the marine environmentally geochemically contaminated conditions by developing protein three dimensional (3D) structures. Foraminifera protein encoding transcripts, attributed primarily to those involved in protein synthesis, intracellular protein trafficking, and modification of the cytoskeleton. This shows that many species of microbiota were not only surviving but thriving, under the severe contaminated environments. The microbiota is characterized by amino acids, fermentation of sugars, fumarate reduction, proteins, nitrate reduction. In addition, the generated data show that under polluted situation Foraminifera may use the creatine phosphate as an Adenosine triphosphate (ATP) store, allowing reserves of high-energy phosphate pool to be maintained for sudden demands of increased energy during anaerobic metabolism. This was co-expressed alongside genes involved in phagocytosis and clathrin-mediated endocytosis (CME). Foraminifera may use CME to utilize dissolved organic matter as a carbon and energy source, in addition to ingestion of prey cells via phagocytosis. This dynamic mechanism helps to explain the environmental survival of benthic foraminifera as recorded in the archives of both shells and sediments. This is considered to be first of its kind work, protein 3D structures of benthic foraminifera (*A. dentata*), as a tool to monitor the marine pollution of India Subcontinent. This protein 3D structure is an initiative for augmentation benthic Foraminiferal metatranscriptomics for future investigations from Indian marine environments in specific and across the world in general.

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1. Introduction

Benthic Foraminifera are the most abundant, ubiquitous, cosmopolitan and free-living marine eukaryotes on the earth both in the recent and fossil record since the Cambrian period (Sreenivasulu et al., 2019). It survived all mass extinction events including marine contamination (Altenbach, 2012). Benthic foraminifera inhabit in and on marine sediments where they account over 55% of the biomass in shallow depths of the seafloor and play a key role in the carbon and nitrogen cycles. Benthic Foraminifera are said to be resistant to contamination and may persist in the benthic biomass even under the development of toxic conditions of both hypoxia and anoxia conditions (Gooday et al., 2008). To this end, we employed protein 3D structure analysis of benthic Foraminifera (*Ammonia dentate)* from coastal sediments of East coast of India. Our analysis showed the pathways of ATP production, and revealed the biosynthetic processes that consume ATP. The data indicate that some Foraminifera affiliated with the genera *Ammonia* are not only surviving under contaminated conditions, but that their transcriptional and cellular activity is not hampered by pollution (Alve et al., 2019). Benthic Foraminifera produce adequate energy to power a multitude of energetically expensive cellular processes in the absence of oxygen, if the environment is highly polluted (Orsi et al., 2020a). Benthic Foraminifera are not only surviving under adverse pollution but appears to thrive under contaminated conditions as archived in the protein3D structures generated in this study.

2. Study area

The study area lies between 80°19′0″–81°23′ 0″ E and 15°17′30″–15°22′30″ *N* (Fig. 1). Earlier researchers have made scientific contributions with different goals such as environmental study (Sreenivasulu et al., 2017; Jayaraju et al., 2021; Lakshmanna et al., 2022). The area under investigation is characterized by tropical to sub-humid conditions with an average temperature of 30 °C. Temperatures during summer reach 31 °C, while in winter fell to 25 °C. The humidity ranges from 71 to 73% in summer to 88 to 90% in winter, resulting in a relatively faster rate of evaporation. Microorganisms are believed to share a common biological process of the existing marine environment.

3. Methodology

Samples collected were transferred immediately into sterile falcon tubes and then refrigerated until further DNA extractions. Sediment samples were washed over a mesh size of 63-micron. The residue, then obtained was sorted to isolate and separate foraminifera tests from sediment grains. The specimens then were identified, counted and catalogued to a species level by adopting standard classification proposed and enumerated (Loeblich and Tappan, 1987). Representative specimens were microphotographed using a CANNON Camera attached Computer. The Ribonucleic acid (RNA) was recovered from sizeable quantity (0.5 g) of sediment by adopting the standard protocols following the manufacturer's manual suggestions and instructions with final elution of templates in 40 μL PCR water (Roche) to increase RNA yield and retard DNA contamination (Orsi et al., 2020b). The benthic foraminifera ORFs assigned were catalogued and annotated suitably. The lack of metatranscriptomic ORFs having highest similarity to *Ammonia dentata* is generally explained by the lack of gene sequencing data from the prototypes of these species in data banks. However, as we are not sure from which foraminiferal taxon each of ORF derives, all of the ORFs have been annotated which have the highest homology to a earlier sequenced genome, as generated from benthic Foraminifera. Clusters of contaminating organisms were recognized by the gene sequencing of 18S rRNA. Contamination was usually associated with sediment bacterial genera including *Acinetobacteria, Burkholderia, Pseudomonas, Ralstonia, Rhizobium, Staphylococcus, Treptococcus*, etc., which are common contaminants (Orsi et al., 2020a).

4. *In silico* investigations

The required calculations were executed on a workstation AMD Opteron Dual-Core 2.0 GHz and 4 GB RAM. However, molecular modelling was carried out using Modeller11v2 (Pawlowski and Holzmann, 2014). We have performed DNA extraction and PCR on *A. dentata*. The expected 18S rRNA gene PCR product size is approximately 1.8–2 kb. For this species no distinct target band was observed at size 1.8–2 kb. From our samples, Gene FASTA sequence submitted to NCBI Open Reading Frame Viewer tool, we found 38 ORFs from which we found 22 sequences have templates and we build 3D structure for these 22 sequences (Orsi et al., 2020a). The details of protein sequences and it related templates are given (Table 1). The Modeller 11v2 software was used for protein modelling and PyMol software for protein structure visualization (Figs. 6 to 8). The broad protocol adopted in this work is given in the form of a flow chart (Fig. 2).

Species identification:

The following classification was adopted for benthic foraminifera (*Ammonia dentata)* under investigation from the present study.

Classification

Phylum: PROTOZOA
Subphylum: SARCODINA
Order: FORAMINIFERIDA
Suborder: ROTALIINA
Species: *Ammonia dentate* (Tappan and Loeblich, 1968), (Fig. 3A, B)
Asterorotalia dentata (Parker and Jones, 1865)

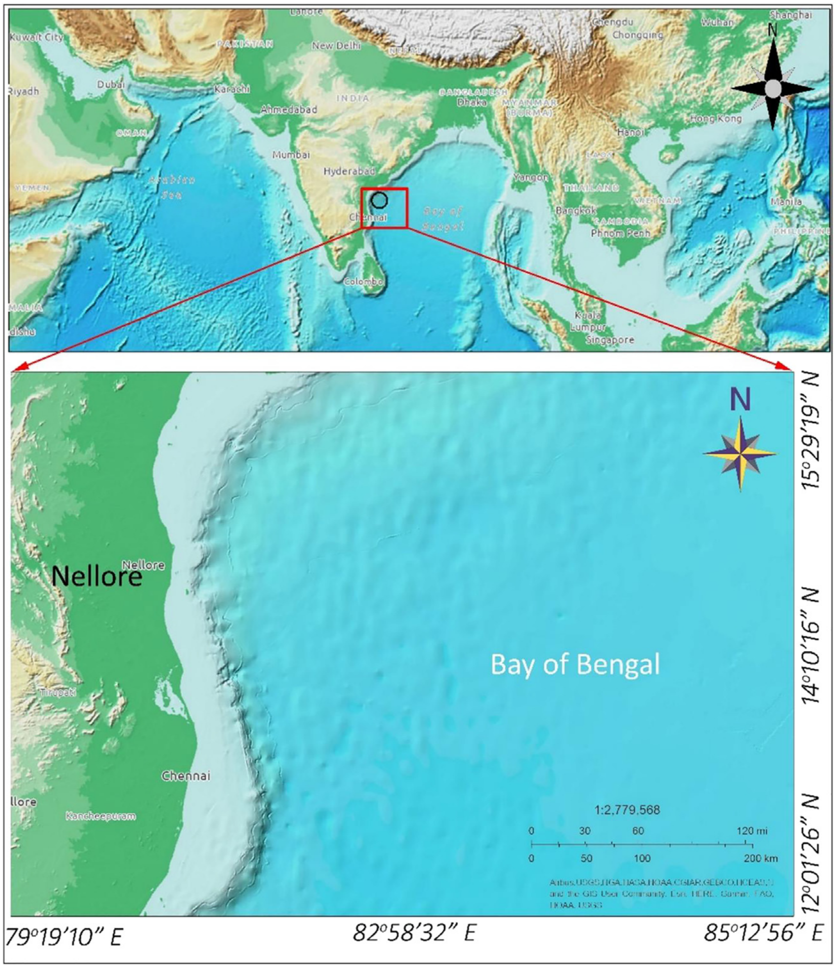

Fig. 1. Location map of the study area.

5. Results and discussion

Benthic Foraminifera shells were studied under microscope which shows decrease in abundance with increase in pollution intensity. However, foraminiferal microbiota, still occur in the polluted area indicating that fauna was thriving and surviving under adverse contaminated ecosystems (Fig. 3C). Usually, the foraminiferal tests posses' cytoplasm, inferring that they were alive. However, presence of oxygen was detected below the sediment surface. Relatively, the gene sequencing level of benthic Foraminifera decreased with increase in pollution intensity. This may be attributed to the total number of expressed protein encoding ORFs assigned to Eukaryotes decreased (Figs. 6–8). Many ORFs of *A. dentata* cannot be explained by a reduction in metatranscriptomes from other several hemolytic communities.

Phylogenetic analyses of Foraminifera 18S rRNA sequences had strong affinity with earlier documented Foraminiferal 18S rRNA sequences. Abundance of *A. dentata* gradually decreases with increase in pollution. This species was dominant and abundant taxon identified (Fig. 4). The detection of its 18S rRNA gene sequencing confirms that this integrated approach captured the activity of this abundant and most opportunistic species. In contrast to 18S rRNA sequences, the foraminiferal ORFs has more homology with earlier sequenced genomes and transcriptomes of *Ammonia, Elphidium, Globobulimina* and *Rosalina* cells (Orsi et al., 2020a). The genome or transcriptome data for *A. dentata* was not available publicly to compare with our datasets for cataloguing and annotating the gene sequencing data. Therefore, that only 18S rRNA was detected from *A. dentata* (Fig. 5). The gene sequencing data of *A. dentate* infer the following mechanisms for the ATP production. They are (i) substrate level amino acids and phosphorylation of sugars by means of fermentation and glycolysis (ii) Application of fumarate as a terminal electron acceptor through fumarate and (iii) reduction of nitrite to produce protons for generation of ATP by ATP syntheses (Figs. 6–8). The ORFs encoding F-actin proteins have associated in the process of biomineralization of shells made up of with Calcium Carbonate. The ORFs of *A. dentata* with homology to protein diaphanous homolog 1 were also recorded (Fig. 5a) (Tyszka et al., 2019). This is considered to be responsible of the

Table 1
Protein sequences and it related templates.

S.no	Protein name	Template (pdb ID)	Reference
1	ORF 1 *A. dentata* **E18F**	3gnn.1.A.pdb	SSGCID (2009)
2	ORF 2 *A. dentata* **E18F**	4cgu.1.B.pdb	Pal et al. (2014)
3	ORF 3 *A. dentata* **E1772R**	4eiw.1.A.pdb	Suno et al. (2006)
4	ORF 7 *A. dentata* **E18F**	4ypj.1.A.pdb	Ishikawa et al. (2015)
5	ORF 8 *A. dentata* **E18F**	4rnx.1.A.pdb	Daugherty et al. (2015)
6	ORF 9 *A. dentata* **E1772R**	6r7l.1.B.pdb	Kater et al. (2019)
7	ORF 10 *A. dentata* **E1772R**	4et0.1.A.pdb	Li et al. (2012)
8	ORF 11 *A. dentata* **E18F**	4z8l.1.A.pdb	Wu et al. (2015)
9	ORF 12 *A. dentata* **E18F**	2jyt.1.A.pdb	Tolkatchev et al. (2008)
10	ORF 13 *A. dentata* **E18F**	7jtk.1.a.pdb	Gui et al. (2021)
11	ORF 14 *A. dentata* **E1772R**	4v1a.1.B.pdb	Greber et al. (2014)
12	ORF 15 *A. dentata* **E1772R**	3qjo.1.A.pdb	Jaenicke et al. (2011)
13	ORF 16 *A. dentata* **E18F**	5fu6.1.B.pdb	Raisch et al. (2016)
14	ORF 18 *A. dentata* **E18F**	4fc8.1.A.pdb	Xiang et al. (2012)
15	ORF 19 *A. dentata* **E18F**	7me6.1.A.pdb	Damle et al. (2021)
16	ORF 21 *A. dentata* **E18F**	5tcu.1.O.pdb	Belousoff et al. (2017)
17	ORF 22 *A. dentata* **E18F**	7nbv.1.A.pdb	Hill et al. (2021)
18	ORF 23 *A. dentata* **E18F**	2hjj.1.A.pdb	Swapna et al. (2006)
19	ORF 24 *A. dentata* **E18F**	7qdw.1.A.pdb	Chagot et al. (2022)
20	ORF 25 *A. dentata* **E1772R**	6nf8.1.H.pdb	Koripella et al. (2019)
21	ORF 30 *A. dentata* **E18F**	6pfl.1.C.pdb	Halabelian et al. (2019)
22	ORF 31 *A. dentata* **E1772R**	2cjs.1.A.pdb	Lu et al. (2006)

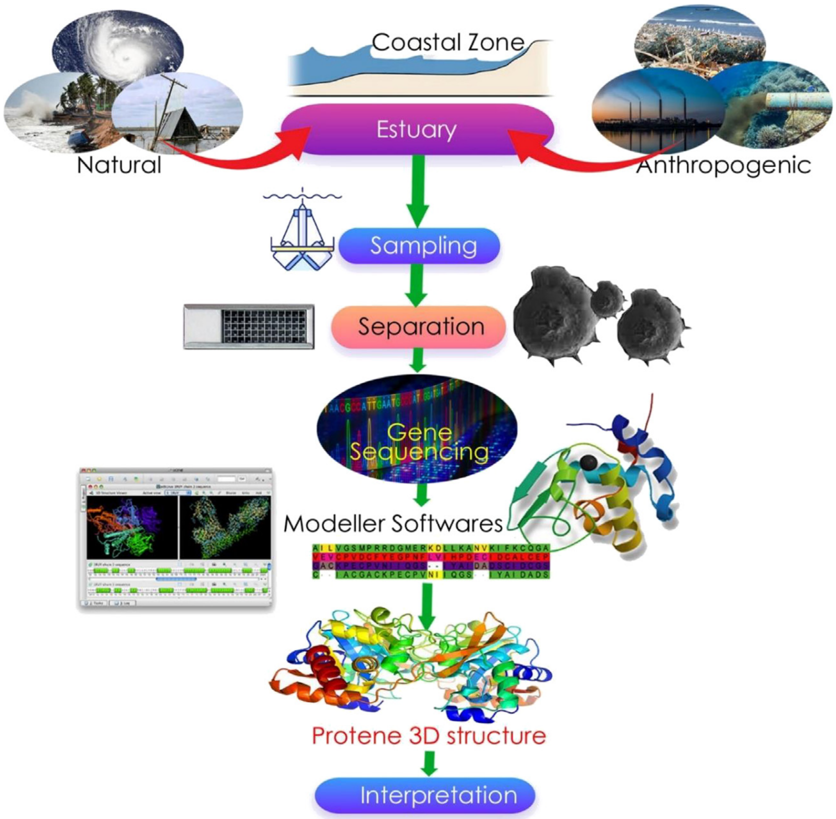

Fig. 2. Flow chart of the protocol for the Protein 3D structures of Benthic foraminifera.

environmental instinct and reason for F-actin nucleation and elongation gradient required for F-actin protein structures. This helps to speculate that these results are an indication that the *A. dentata* were using the encoded Rho proteins and vacuolar-type H+ ATPases proteins to carry out a phagocytosis under contaminated marine ecosystem (Table 1; Figs. 6–8). The derived ORFs were also indicated that encoded dynein, kinesin and tubulins, are probably responsible for sending and receiving cellular cargo to and from the membrane, respectively. However, myosin motor proteins, in addition to

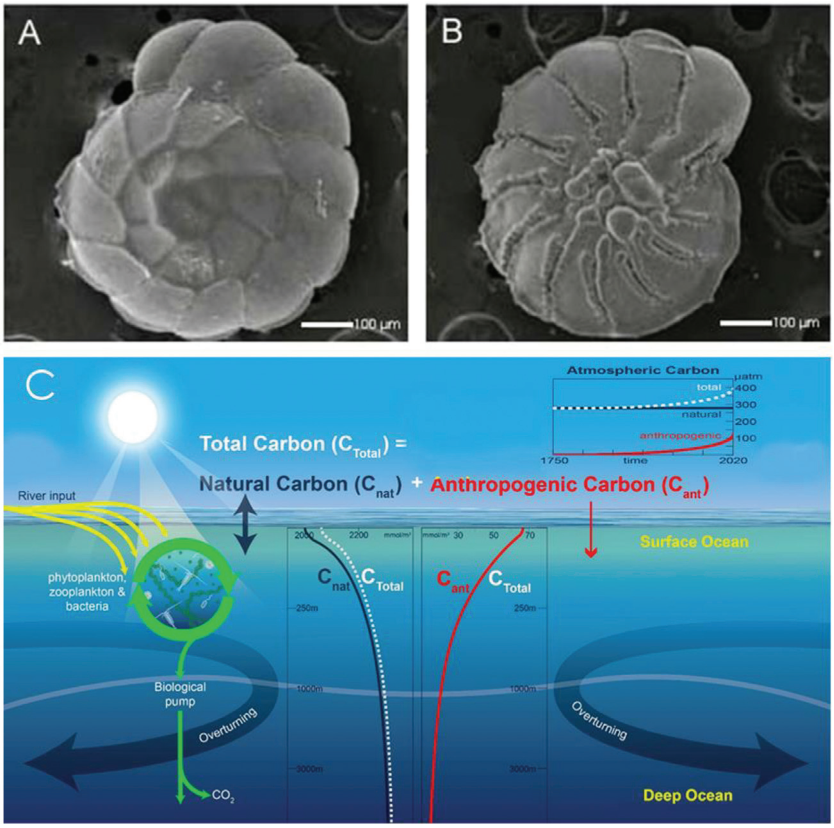

Fig. 3. (A) *Ammonia dentata* dorsal view; (B) *A. dentata*ventral view; (C) A conceptual model for total carbon dynamics in marine ecosystem.

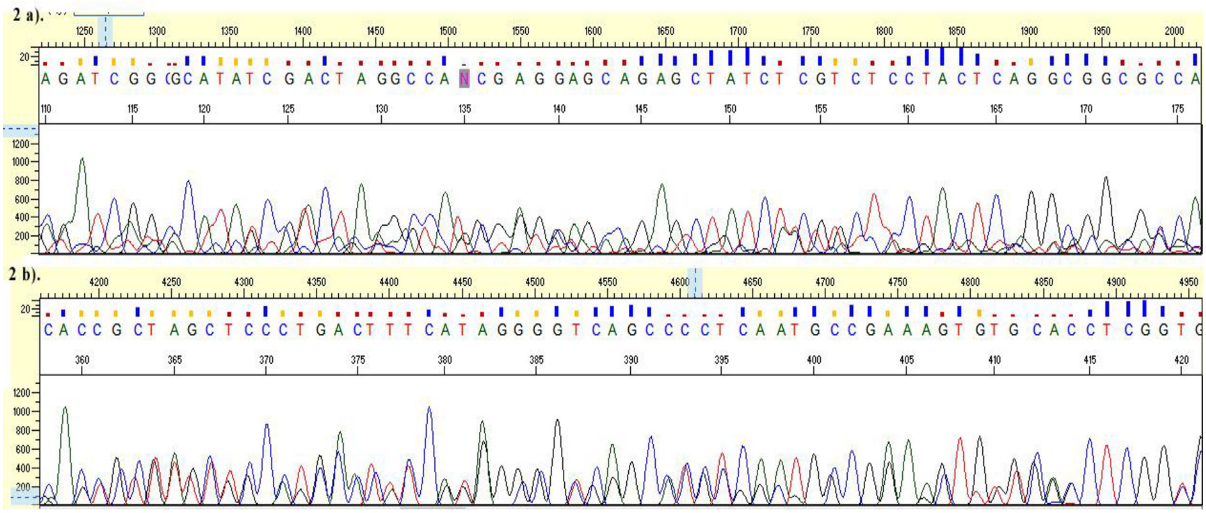

Fig. 4. (a) Representative Eukaryote DNA Barcoding 1st Base4391422 *A. dentata*_1_E1772R.ab1 the Coordinates (*x*, *y*) 1265, 1357 and the Nucleotide range is 110 to 176. 2 (b) Representative Eukaryote DNA Barcoding 1st Base4391421 *A. dentata*_1_E18F.ab1 the Coordinates (*x*, *y*) 4611, 75 and the Nucleotide range is 358 to 421.

phagocytosis play a vital role in many cytoskeletal mechanisms viz., cell adhesion, cell division and cell migration (Tsai and Discher, 2008).

In addition, it composed of 18S rRNA in full sequencing (Fig. 4) from the dominant and cosmopolitan taxon, i.e., *A. dentata*. and thus, it is evident that the ORFs are basically obtained from cytoplasm-containing benthic microbiota.

(a) Gene FASTA Sequence of 1st_BASE_4391422_A.dentata1_E1772R: 1 to363

G T T T C T T GG A GC T A T T G A T A A T G T T T T T T T C T T GG T G A T C A C
T GCG A C A T N C A C A A A G AG A GG A G T A T CG A C C GC A CG A T G G T
C A CGC T G A T C A T CG T G A CGC CGC A T G AG A T CGGCGC A T A T C G A
C T A GGC C A N C G A GG A GC A G AGC T A T C T C G T C T C C T A C T C A
GGC GGC GC C AGC C T C T CG T C C C A G A C C C A C A A G A T C A T C G A C
G A A G A C GC T CGC CGG A T C A T C G A C A A GGGC T A C G A G A C C GC C
A A GCG T A T C T T G A N C G A GCGGCGC G A GG A T C T C G A GCG T C T
GGCGC A T GGC C T C C T C G A G T A C G A G A CGC T G A C C GGC A A C G T
A G A T C A T C CGCG T G A T AG T CGGCG A A C C GC

(b) Gene FASTA Sequence of 1st_BASE_4391421_A.dentata_1_E18F : 1 to 1136

G T GG A T CG A C A A T G T T C T G A T GC A C C GGC A A G T G T T T GC C T
GCG T C T GCG T T C G T C C A T G A T G T C T CGCG A C T A CG T G A C G CGC
C T C T G T C T C A A CGC C A T C T C C T A C A T G T G T C C T A C A T C T A C
C T C CGGC C T T CGGG T AGG A A A A T C C T G A C A T CGG A GC A T G A C T
T C GC A T C T A CG A C A AGGCG A T G A GCGC T G A A GC A C CGC G A G T
GGC A C C T GCG T C CGC A A A C T T G T T C T GC C AG T G C C A C T GG T A C
A CGC A C C T C C T GC T GG T T C T A T A T C GGC T CG T GC T G T GGGGG A
GG A G A GG T A C C T T T A T C T CG A AGGGCGC A A CGGCG A A C A CGC T A
T C C T T C A GG A T A C CGG A C A C CGC T A GC T C C C T G A C T T T C A T A
GGGG T C A GC C C C T C A A T GC CG A A A G T G T GC A C C T CGG T GGCGG
T A GC C T C T C C T C C A C CG A G A C A C C T GG T A C T T C GGC AG A A T G
AGG T A A C T C G A A C A T G A A C A A C A T C C T GGC T G A T CGG T T
GCGCGC T GC C T C T C T C T A G T C T T G A C C A CGCGGC CGC T T G A C T A
C A T GC T C T C C T T C A T T G A C T GGC T G AGGGCGCGGGGC T T T T T GC T
G A T C GCG A T C T T C G T C GC A T A C T T C C T G AGG A T C GGC C T T T
GGG A GG A A GCGC T C G T GC T G T G A C A A GG T A T C T A G A T C T A C C
T C T A C G T T A T C AGC C A GC C T A A C CG T C C A C T A T T G T C A T T A T
CGC T C A A G T T G AGC T GC G T A C C C T GC T C A T CG T C A A T C C T T GC
A A C C GC A C A A T G A T C T C A C GN T GCG A A G T T C GC C C T C C A T A
C C G A C G CG T C T GC T A A C C T G A A T C A A CGC C C A G A A GGC C T G T
C T C C C AGC T GC A T A A A G A T T G T A T CGC T C A CG A CG A T C A T GGC
C G C T GG T A T C CGG A A GG T C C T N N A C A T C GGC A C T GC C C A C A T
G A A A C C A T GGCG T A A T AG T T CGCGC T A T GN A T GGGC A G A T T GGC
T A T C T GGC A A A T C T G A A G T G AGC C A C C C A C C A T T C C C C GGGG
A T A C C C C A T GC T GGC T C C A A A A G T C T C C C A T C CG T C C A T
GGGC C C T T G A A T CG T C C N T C G A A A A GGGG T T C C T T GGGC A A A T
G A T C C T A A A GC T T GGGCGGC C C A T C C T T C C T T GGG CG

Fig. 5. (a) Gene FASTA Sequence of 1st_BASE_4391422_*A. dentata*1_E1772R: 1 to 363; (b) Gene FASTA Sequence of 1st_BASE_4391421_*A. dentata*_1_E18F : 1 to 1136.

However, the abundance of *A. dentata* is low in the contaminated zone samples, but relatively abundant with those at the surface of contamination. Obviously, there was a relatively abundant crop of Foraminifera transcripts involved in energy production in the unpolluted zones, in comparison with the polluted areas. This can be speculated that this may be owing to the availability of nutrients at the nexus of sediment and seawater. This perhaps increases aerobic respiration and catalysis energy metabolism of benthic foraminifera in general and *A. dentata* in specific (Hayward et al., 2019).

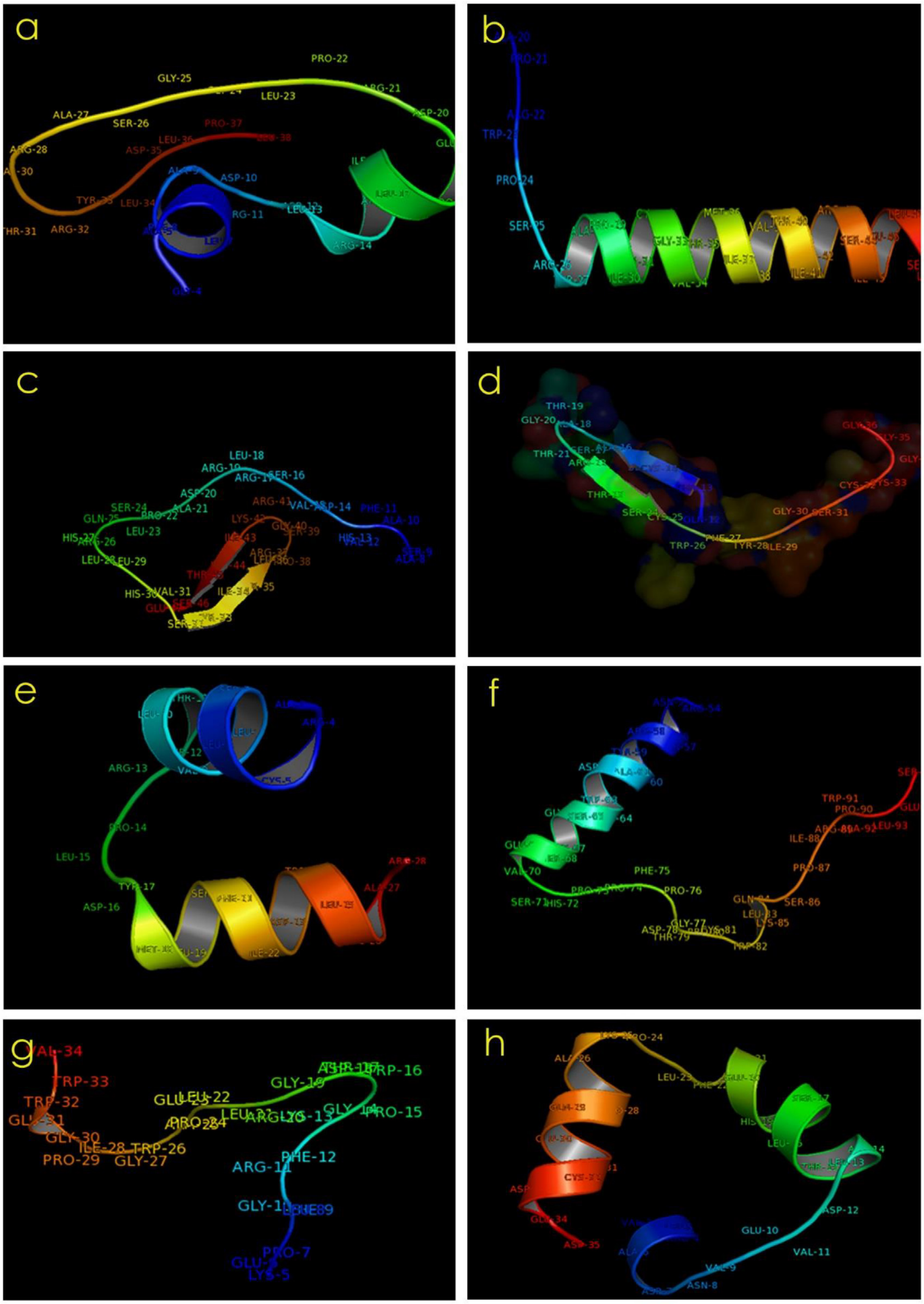

Fig. 6. Developed 3D Structure of (a) ORF 8 ***A. dentate*_E18F**; (b) ORF 9 ***A. dentata*_E1772R**; (c) ORF 11 ***A. dentata*__E18F**; (d) ORF 12 ***A. dentata*_E18F**; (e) ORF 13 ***A. dentata*_E18F**; (f) ORF 15 ***A. dentata*__E1772R**; (g) ORF 16 ***A. dentata*_E18F**; (h) ORF 18 ***A. dentata*_E18F**.

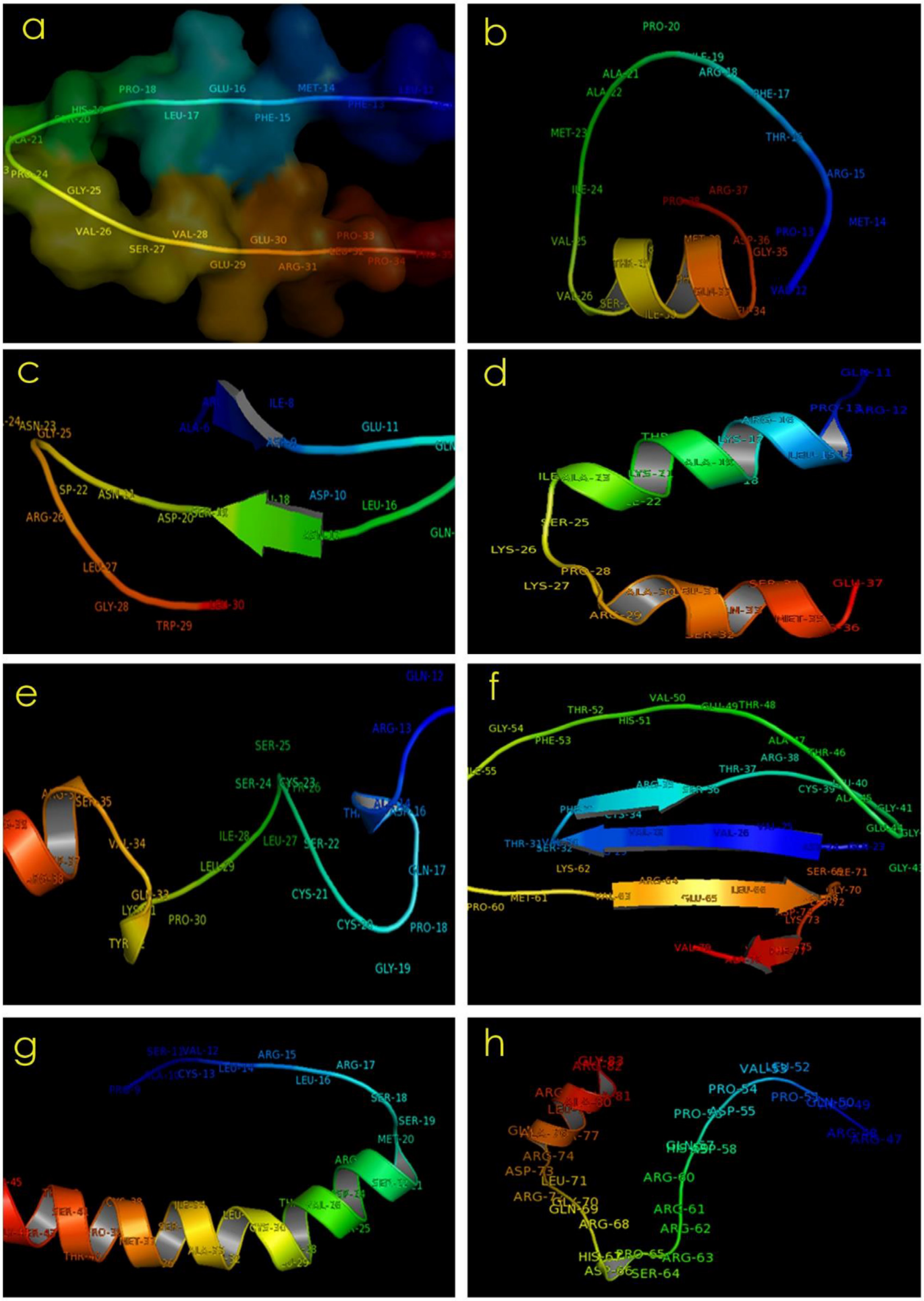

Fig. 7. Developed 3D Structure of (a) ORF 19 ***A. dentata*_E18F**; (b) ORF 22 ***A. dentata*_E18F**; (c) ORF 23 ***A. dentata*_E18F**; (d) ORF 24 ***A. dentata*_E18F**; (e) ORF 25 **A. dentata_E1772R**; (f) ORF 31 ***A. dentata*_E1772R**; (g) ORF 1 ***A. dentata*_E18F**; (h) ORF 1 ***A. dentata*_ E1772R**.

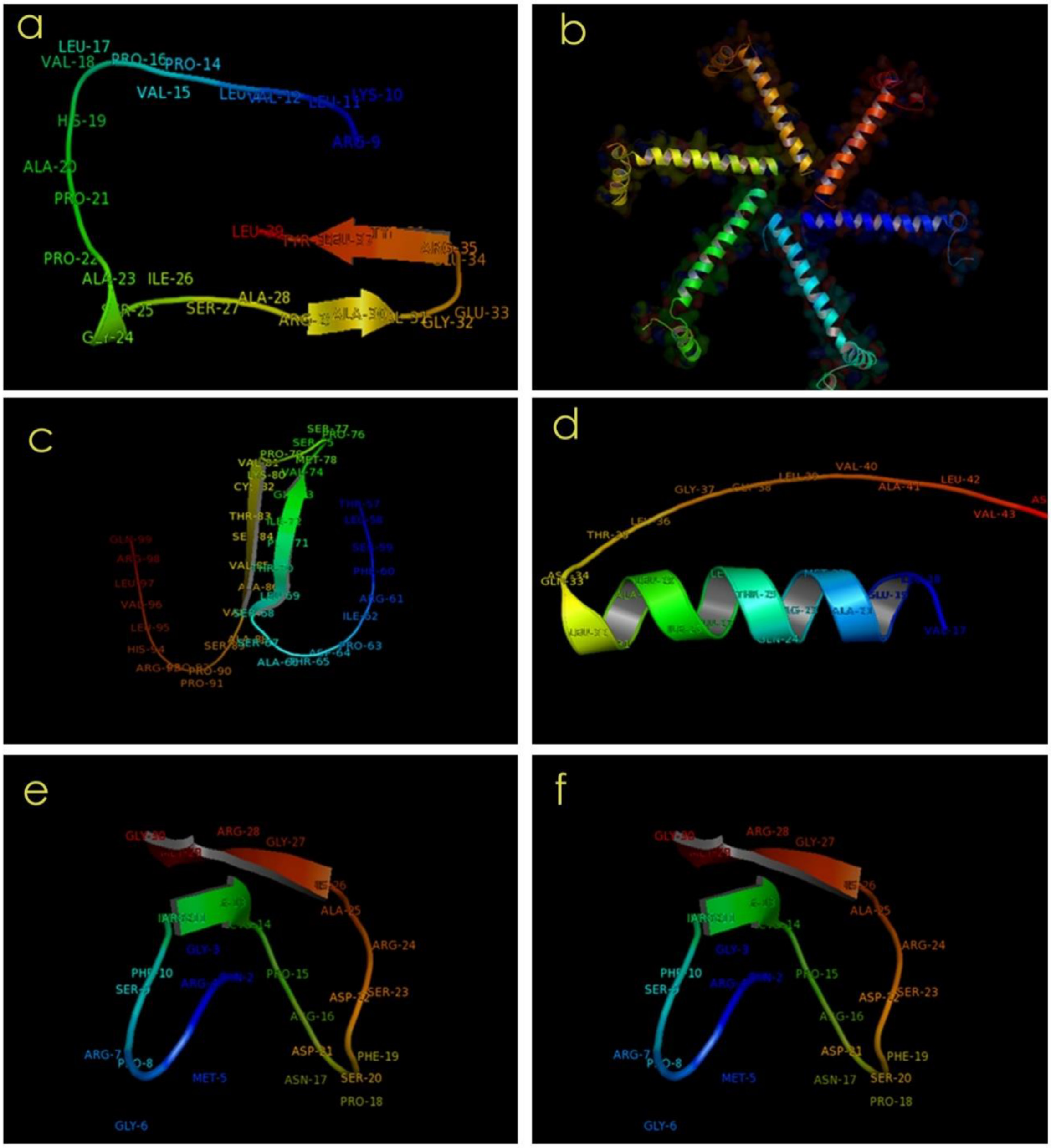

Fig. 8. Developed 3D Structure of (a) ORF 2 ***A. dentata*_E18F**; (b) ORF 3 ***A. dentata*_E1772R**; (c) ORF 7 ***A. dentata*_E18F**; (d) ORF 10 ***A. dentata*_E1772R**; (e) ORF 21 ***A. dentata*_E18F**; (f) ORF 30 ***A. dentata*_E18F**.

The ORFs encoding Rho proteins by *A. dentata* indicates an active role of phagocytosis, since Rho proteins function in actin dynamics during phagocytosis (Doherty and McMahon, 2009). However, myosin motor proteins are absorbed by cell membrane during phagocytosis to enhance and capture prey material as a food vacuole (Bird et al., 2020). The proteins datasets show that under contaminated marine environments, *A. dentata* metabolize the hydrolysed organics for ATP production via fermentation, fumarate and nitrite reduction (Fig. 5). Generally, cells are largely made up of with protein, anaerobic fermentation of ingested prey cells that dissolves amino acid fermentations. The fast growing cells are roughly made up of with about 60% protein, 20% RNA, 10% lipids, 3% DNA, about 20% sugars and some insignificant amounts of metabolites (Orsi et al., 2020a). During respiration in marine eukaryotes malate is translated in to fumarate via the fumarase enzyme. The resultant fumarate can then serve as the terminal electron acceptor (Müller et al., 2012). This process is widely noticed amongst eukaryotes viz., *Bivalvia, Ciliophora, Euglenida, Foraminifera, Nematoda, Polychaeta, Platyhelminthes* (Müller et al., 2012).

Out of 31 ORFs generated, only 22 ORFs of *A. dentata* were recorded other missing ORF include ORFs 4, 5.6,17,20,26,27,28 and 29. Missing ORFs might have truncated owing to contamination (Martin et al., 2017). This indicate that fauna increased the number of expressed genes, rather than the increase owing to the dying off of other marine eukaryotes causing hike in abundance of Foraminifera gene sequencing. The high number of ORFs *A. dentata* may probably be due to those involved in the cytoskeleton, translation, posttranslational modification, and intracellular trafficking (Orsi et al., 2020a, Figs. 6–8). This shows that much crop were changing their physiology, increasing translation and protein synthesis in response to pollution conditions. The present study demonstrates that dynamic activity of *A. dentata* in these contaminated environment is hampered by local conditions, homology of oxygen minimum zone (Pawlowski et al., 2013). The low activity of benthic foraminifera gene metatranscriptomics at the contaminated conditions might indicate the non availability of nutrients such as fumarate, nitrate, nitrite etc.

6. Conclusions

From all the data generated, it is clear that the metabolic acidity of these benthic foraminifera is not adequate to support biocalcification and phagocytosis, and under polluted condition. Creatine kinases and the dephosphorylation of creatine phosphate appears to play a key role in maintaining high-energy levels of phosphate potentially enabling short lived bursts of energetically demanding activities under polluted conditions such as phagocytosis of prey cells. This all shows that contaminated sediments are generally a primary nursery ground for some benthic Foraminifera where they are able to perform all necessary metabolic activity thriving and sustaining in the dubious ecological conditions (Richirt et al., 2021). This is the first report on protein 3D structures of benthic foraminifera (*A. dentata*), as a tool to monitor the marine pollution of India Subcontinent. This protein 3D structure is an initiative for augmentation of benthic Foraminiferal gene sequencing for future investigations from Indian marine environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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